Levels of structural organisation

LEVELS OF STRUCTURAL ORGANISATION (sample notes)

The body is organised in a hierarchy:
1. Chemical level: atoms combine into molecules (for example, proteins, lipids, nucleic acids).
2. Cellular level: molecules form organelles and cells, the smallest living units.
3. Tissue level: groups of similar cells with a common function. The four primary tissue types are epithelial, connective, muscle and nervous tissue.
4. Organ level: two or more tissue types working together (for example, the stomach).
5. Organ system level: organs that cooperate on a common purpose (for example, the digestive system).
6. Organismal level: all organ systems together form the whole organism.

Clinical link: many diseases are understood by asking which level is affected. Atherosclerosis, for example, begins at the cellular and tissue level (endothelial injury, lipid deposition) and becomes an organ-level problem when it narrows coronary arteries.